University of Oklahoma
Department of Mathematics
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MATH 3333–004
Linear Algebra
Spring 2026

MIDTERM EXAM 2 (FORM B)

NAME: SOLUTIONS

ID N°:

INSTRUCTIONS

- This is a closed-book exam. You may not use your notes, textbook or any outside resources to assist you.
- Write your answer to each question in the space assigned to each problem. If you want the instructor to read the scratch working page for the answer to a specific problem, explicitly write that on the space assigned to said problem.
- Always remember to show your work. An answer without a proper justification will earn little to no credit, even if it is correct.
- Circle or box your final answer to each question.
- Make sure that your answers are organized and legible. Use a dark pencil or a pen. Any work that is hard or impossible to read will not be graded.
- When solving a problem, if you are going to use an alternative method to the ones discussed in class (for instance, inverting a matrix via adjugates or using Cramer's rule to solve a linear system), you are required to include a short explanation on how this alternative method works
- The use of calculators of any type is not allowed.

This exam consists of four problems, and your total score will be out of $10 + 10 + 10 + 10 = 40$ points.

Good Luck!

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Problem 1 (10 points)**.** Consider the linear transformation $T\colon \mathbb{R}^2 \to \mathbb{R}^3$ that satisfies

$$T\left(\begin{bmatrix}1\\-2\end{bmatrix}\right) = \begin{bmatrix}7\\1\\0\end{bmatrix} \qquad \text{and} \qquad T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}-2\\0\\1\end{bmatrix}.$$

(a) (6 points) If $\mathbf{e}_1$ and $\mathbf{e}_2$ are the standard basis vectors of $\mathbb{R}^2$, find $T(\mathbf{e}_1)$.

$\vec{e}_1 = \begin{bmatrix}1\\0\end{bmatrix}$, $\vec{e}_2 = \begin{bmatrix}0\\1\end{bmatrix}$. Is $\vec{e}_1$ a linear combination of $\begin{bmatrix}1\\-2\end{bmatrix}$ and $\begin{bmatrix}0\\1\end{bmatrix}$?

$$\begin{bmatrix}1\\0\end{bmatrix} = a\begin{bmatrix}1\\-2\end{bmatrix} + b\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}a\\-2a+b\end{bmatrix} \iff \begin{cases}a=1\\-2a+b=0\end{cases} \Rightarrow \begin{matrix}a=1\\b=2\end{matrix}$$

Therefore, $\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}1\\-2\end{bmatrix} + 2\begin{bmatrix}0\\1\end{bmatrix}$. If we apply T, we obtain

$$T(\vec{e}_1) = T\left(\begin{bmatrix}1\\-2\end{bmatrix} + 2\begin{bmatrix}0\\1\end{bmatrix}\right) \underset{\substack{\uparrow\\ T \text{ linear}}}{=} T\left(\begin{bmatrix}1\\-2\end{bmatrix}\right) + 2T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}7\\1\\0\end{bmatrix} + 2\begin{bmatrix}-2\\0\\1\end{bmatrix} = \boxed{\begin{bmatrix}3\\1\\2\end{bmatrix}}$$

(b) (4 points) Compute the matrix of T.

The matrix of T is

$$A = \begin{bmatrix}T(\vec{e}_1) & T(\vec{e}_2)\end{bmatrix} = \boxed{\begin{bmatrix}3 & -2\\1 & 0\\2 & 1\end{bmatrix}}$$

Problem 2 (10 points). Answer the following questions concerning elementary matrices:

(a) (5 points) Consider the matrix

$$A = \begin{bmatrix} 1 & -2 & -2 \\ 2 & -1 & -1 \end{bmatrix}.$$

Find the **reduced** row-echelon form R of A and an invertible matrix U such that $UA = R$.

$$\left[\begin{array}{ccc|cc} 1 & -2 & -2 & 1 & 0 \\ 2 & -1 & -1 & 0 & 1 \end{array}\right] \xrightarrow[R_2-2R_1]{} \left[\begin{array}{ccc|cc} 1 & -2 & -2 & 1 & 0 \\ 0 & 3 & 3 & -2 & 1 \end{array}\right] \xrightarrow[\frac{1}{3}R_2]{} \left[\begin{array}{ccc|cc} 1 & -2 & -2 & 1 & 0 \\ 0 & 1 & 1 & -2/3 & 1/3 \end{array}\right] \xrightarrow[R_1+2R_2]{} \left[\begin{array}{ccc|cc} 1 & 0 & 0 & -1/3 & 2/3 \\ 0 & 1 & 1 & -2/3 & 1/3 \end{array}\right]$$

The reduced row-echelon form of A is $\boxed{R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}}$ and the matrix $\boxed{U = \frac{1}{3}\begin{bmatrix} -1 & 2 \\ -2 & 1 \end{bmatrix}}$ is an invertible matrix satisfying $UA = R$.

(b) (5 points) Write the matrix

$$B = \begin{bmatrix} 1 & 4 \\ 2 & 7 \end{bmatrix}$$

as a product of elementary matrices.

$$\left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array}\right] \xrightarrow[R_2-2R_1]{} \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 0 & -1 & -2 & 1 \end{array}\right] \xrightarrow[-R_2]{} \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 0 & 1 & 2 & -1 \end{array}\right] \xrightarrow[R_1-4R_2]{} \left[\begin{array}{cc|cc} 1 & 0 & -7 & 4 \\ 0 & 1 & 2 & -1 \end{array}\right]$$

We get that $B^{-1} = E_{R_1-4R_2} E_{-R_2} E_{R_2-2R_1}$, so taking inverses we obtain

$$\boxed{B = E_{R_2+2R_1} E_{-R_2} E_{R_1+4R_2} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}}$$

Problem 3 (10 points)**.** Solve the following problems related to determinants.

(a) (5 points) Assuming $\begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix} = -1$, calculate $\begin{vmatrix} -2a & -2b & -2c \\ 2p+x & 2q+y & 2r+z \\ 3x & 3y & 3z \end{vmatrix}$.

$$\begin{vmatrix} -2a & -2b & -2c \\ 2p+x & 2q+y & 2r+z \\ 3x & 3y & 3z \end{vmatrix} = (-2)\cdot 3 \begin{vmatrix} a & b & c \\ 2p+x & 2q+y & 2r+z \\ x & y & z \end{vmatrix} \underset{R_2-R_3}{=} -6 \begin{vmatrix} a & b & c \\ 2p & 2q & 2r \\ x & y & z \end{vmatrix}$$

$$= -6\cdot 2\cdot \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix} = -6\cdot 2\cdot (-1) = \boxed{12}$$

(b) (5 points) Find the values of x for which the matrix

$$A = \begin{bmatrix} 2 & 1 & -2 \\ x & -2 & 1 \\ 7 & 0 & -x \end{bmatrix}$$

is **not** invertible.

The matrix A is invertible if and only if $\det(A) = 0$. We compute $\det(A)$:

$$\begin{vmatrix} 2 & 1 & -2 \\ x & -2 & 1 \\ 7 & 0 & -x \end{vmatrix} \underset{R_2+2R_1}{=} \begin{vmatrix} 2 & 1 & -2 \\ x+4 & 0 & -3 \\ 7 & 0 & -x \end{vmatrix} = -\begin{vmatrix} x+4 & -3 \\ 7 & -x \end{vmatrix} = -\left((x+4)(-x)+21\right) = x^2+4x-21 = (x-3)(x+7)$$

We conclude that:

$$\boxed{A \text{ is not invertible} \iff \begin{matrix} x=3 \\ \text{or} \\ x=-7 \end{matrix}}$$

Problem 4 (2 points per part)**.** Decide whether each of the statements below is true or false. Completely fill the circle that is to the left of your choice. You do not need to justify your choices.

(a) The matrix of the linear transformation $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x - y \\ x + 3y \end{bmatrix} \qquad \text{is} \qquad \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}.$$

● True ○ False

(b) If A is a 4×4 matrix, then $\det(3A) = 81 \det(A)$.

● True ○ False

(c) The determinant

$$\begin{vmatrix} 0 & 0 & a \\ 0 & b & c \\ d & e & f \end{vmatrix} = abd.$$

○ True ● False

(d) The product of two elementary matrices is also an elementary matrix.

○ True ● False

(e) If A is a square matrix and B is the matrix obtained by interchanging two different rows of A, then $\det(B) = \det(A)$.

○ True ● False

Scratch working page

Scratch working page